

Deliverable: Draft an instructional strategy for a sample lesson that introduces teachers to a constructivist approach to either science teaching or, if you currently are a teacher, a sample lesson in your subject area. Be sure to cite at least (2) outside sources.

To define new constructivist lesson plans, I think it is important for us as educators to know who the learners are. In addition, I think that constructivist teachers need a theoretical framework for teaching and learning—an understanding of how people learn and how to apply instructional design strategies. This means that the teacher needs to know what it means to be a constructivist teacher.

According to the theory of constructivism, the role of constructivist teachers is to facilitate learning by assisting learners, and it means encouraging and supporting them (see Constructivism). It is also suggested that learners construct knowledge. This kind of approach promotes learning by doing, enabling learners to construct new knowledge from their experience. Constructivist approaches describe how learning should occur, regardless of whether learners use their experiences to construct knowledge.

According to constructivism, learning is based on tasks (p. 23). I want to emphasize the importance of these tasks. Here is the list:

- 1) provide multiple representations of reality and the complexity of the real world
- 2) present authentic tasks that encourage conceptualization
- 3) provide real-world, case-based contexts
- 4) support deep thinking
- 5) often involve student collaboration featuring social negotiation and cooperative learning structures
- 6) support students to choose from a variety of possible solutions or approaches to a problem
- 7) enable learners to make connections across disciplines and perspectives

I think that these points encourage constructivist teachers to believe in what they teach.

In an article by Hsiao-Ching She (2004), she states three problems concerning science teachers. Even though this article is specifically geared toward science teachers, I think that these can be applied to all other disciplines as well.

1) What types of new perspectives, theories, and instructional strategies about science teaching and learning are needed for science method courses, and how can we make all the prospective science teachers really believe in what they have learned in science method courses?

2) How can we encourage prospective teachers to implement what they have learned in classes during their teaching practice?

3) How can we bridge science method courses and teaching practice, that is, how can we make prospective teachers' beliefs and actions coherent?

Another good article by Chen (2007) describes instruction design principles in online courses. I think this is a good article for anyone planning to teach online. Chen provides a useful comparison between objectivist and constructivist learning and instruction. His conceptual framework is very much in line with constructivist thinking (see pg. 4).

Another theoretical framework in our text states that teacher education must help teachers learn and apply important ideas about teaching (p. 3).

There are two components:

- 1) the input of theoretical ideas and new teaching suggestions, and
- 2) trying out and practicing these over an extended period and in a collaborative situation where the pre-service teachers can receive support and feedback, critically reflect, as well as renegotiate and reconstruct the aims and purposes of teaching.

I believe that I am a constructivist teacher. It's hard to be objectivist in learning and instruction with Computer Science courses. I teach computer programming courses where my instructional strategy primarily involves group discussion, hands-on activities, consultation, and collaboration.

Here is a list of links that are pretty much in line with my constructivist thinking.

Sue-Jen Chen (2007), Instructional Design Strategies for Intensive Online Courses: An Objectivist-Constructivist Blended Approach from <http://www.ncolr.org/jiol/issues/PDF/6.1.6.pdf>

Investigating Multiple Pedagogical Perspectives through MIMIC (Multiple Intelligent Mentors Instructing Collaboratively) from <http://www.hcrc.ed.ac.uk/aied2001/baylor.pdf>

Hsiao-Ching She (2004), Reforming the education program for physical science teachers from http://www.acer.edu.au/documents/AJE_sample_vol48_1f.pdf

Constructivism (learning theory) from [http://en.wikipedia.org/wiki/Constructivism_\(learning_theory\)](http://en.wikipedia.org/wiki/Constructivism_(learning_theory))

An Instructional Approach to Teaching the Research Process based on the Principles of Constructionist and Metacognition from <http://www.coe.uga.edu/epltt/LessonPlans/LPMetacog2LAdams.htm>

Dr. Kim Response
Hi Jim,

Very well done this week with your discussion board posts and your Week 2 Deliverable.

I especially enjoyed reading what you wrote here:

Another theoretical framework in our text states that teacher education must assist teachers to learn and apply important ideas about teaching (pg 3).

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- 1) the input of theoretical ideas and new teaching suggestions, and
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I agree that it is very important to not only understand the cognitive aspect of a topic, but it is also very important to try it out in practice and work with colleagues/peers in a collaborative nature to help enrich the learning experience.

I also liked where you said that it is very important for the teacher to know their students.

Without having an understanding of student needs, it makes the job for a teacher very difficult. With an online class this is especially true. Can you think of some ways that as an online teacher you can find out information about your students that will help guide you as their instructor?

Thank you for all of your hard work this week Jim. It has been a pleasure reading your posts.

Score for Week 2: 10/10

~Dr. Butler